



F323 Clutch Pedal Force Loadcell

The F323 is a compact, high accuracy low profile loadcell for measuring clutch pedal application forces.

The loadcell is robust in construction with a maximum force range of 1kN (100kgf). Its multi-hole fixing allows simple attachment to any shape pedal using screws or cable ties. The low profile design maintains the clutch pedals ergonomic geometry. Low mass aluminium alloy construction reduces the mass influence upon the clutch pedal in dynamic testing.

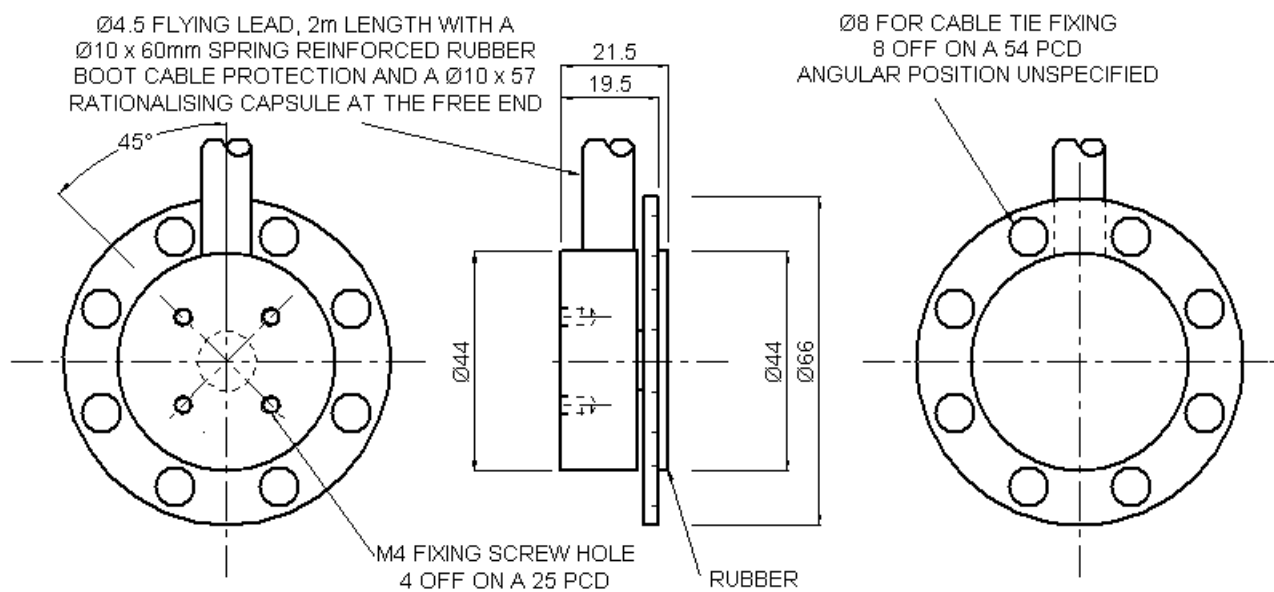
If you require other force ranges the F304 or F322 may be suitable.

We are happy to design variants of this loadcell to meet your specific requirements.

Features

- ✓ Off centre loading compensation
- ✓ Simple cable tie or screw fixing
- ✓ Anti-slip rubber pad
- ✓ Removable loading plate
- ✓ Output rationalised to 2mV/V
- ✓ Lightweight aluminium alloy construction
- ✓ Traceable calibration with certificate included in the standard price

Outline



Specifications

Parameter	Value	Unit
Non-linearity - Terminal	±0.05	% RL
Hysteresis	±0.05	% RL
Creep - 20 minutes	±0.05	% AL
Repeatability	±0.02	% RL
Rated output - Rationalised	2.0	mV/V
Rationalisation tolerance	±0.1	% RL
Zero load output	±10	% RL
Temperature effect on rated output per °C	±0.005	% AL
Temperature effect on zero load output per °C	±0.005	% RL
Temperature range - Compensated	-10 to +50	°C
Temperature range - Safe	-10 to +80	°C
Excitation voltage - Recommended	10	V
Excitation voltage - Maximum	20	V
Bridge resistance	700	Ω
Insulation resistance - Minimum at 50Vdc	500	MΩ
Inclined load error - concentric at 3°	±0.25	% RL
Structural stiffness	1.5 x 10 ⁷	N/m
Overload - Safe	50	% RL
Overload - Ultimate	200	% RL
Sideload - Safe	100	% RL
Sealing	IP65	
Weight - Nominal (excluding cable)	80	g

Order Codes

Code	Description
F323CFR0KN	Compression, IP65, rationalised

Notes

- AL = Applied load.
- RL = Rated load.
- Temperature coefficients apply over the compensated range.

Connections

The loadcell is fitted with 2 metres of flexible polyurethane jacketed 4 core screened cable.

Excitation + = Red, Excitation - = Blue, Signal + = Yellow, Signal - = Green, Screen = Orange.

The screen is not connected to the loadcell body.

When this loadcell is rationalised the resistors are housed in a capsule located in the loadcell cable 100mm from the free end. Capsule dimensions are Ø10mm by 57mm.